```
import pandas as pd
df = pd.read_csv("Training.csv")
df.head()
```

	abdominal_pain	abnormal_menstruation	acidity	acute_liver_failure	altered_sensorium	anxiety
0	0	0	0	0	0	0
1	0	0	0	0	0	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0

5 rows × 135 columns

```
from google.colab import files
uploaded = files.upload()
```

Choose Files 5 files
Healthcare.csv(text/csv) - 2106555 bytes, last modified: 3/15/2026 - 100% done
Symptom-severity.csv(text/csv) - 2332 bytes, last modified: 3/15/2026 - 100% done
dataset.csv(text/csv) - 52229 bytes, last modified: 3/15/2026 - 100% done
Training.csv(text/csv) - 88509 bytes, last modified: 3/15/2026 - 100% done
Testing.csv(text/csv) - 13952 bytes, last modified: 3/15/2026 - 100% done

```
Saving Healthcare.csv to Healthcare (4).csv
Saving Symptom-severity.csv to Symptom-severity (4).csv
Saving dataset.csv to dataset (4).csv
Saving Training.csv to Training (4).csv
Saving Testing.csv to Testing (6).csv
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("dataset.csv")
df.head()
```

	Disease	Symptom_0	Symptom_1	Symptom_2	Symptom_3	Symptom_4	Sy
0	AIDS	muscle_wasting	patches_in_throat	high_fever	extra_marital_contacts	NaN	
1	AIDS	patches_in_throat	high_fever	extra_marital_contacts	NaN	NaN	
2	AIDS	muscle_wasting	high_fever	extra_marital_contacts	NaN	NaN	
3	AIDS	muscle_wasting	patches_in_throat	extra_marital_contacts	NaN	NaN	
4	AIDS	muscle_wasting	patches_in_throat	high_fever	NaN	NaN	

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 313 entries, 0 to 312
Data columns (total 18 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Disease     313 non-null    object
 1   Symptom_0   313 non-null    object
 2   Symptom_1   313 non-null    object
 3   Symptom_2   313 non-null    object
 4   Symptom_3   281 non-null    object
 5   Symptom_4   234 non-null    object
 6   Symptom_5   186 non-null    object
 7   Symptom_6   158 non-null    object
 8   Symptom_7   140 non-null    object
 9   Symptom_8   120 non-null    object
 10  Symptom_9   110 non-null    object
 11  Symptom_10  68 non-null     object
 12  Symptom_11  47 non-null     object
 13  Symptom_12  30 non-null     object
 14  Symptom_13  19 non-null     object
 15  Symptom_14  18 non-null     object
 16  Symptom_15  10 non-null     object
 17  Symptom_16  1 non-null      object
dtypes: object(18)
memory usage: 44.1+ KB
```

	Disease	Symptom_0	Symptom_1	Symptom_2	Symptom_3	Symptom_4	Symptom_5	Symptom_6	
count	313	313	313	313	281	234	186	158	
unique	42	38	51	56	54	38	32	26	
top	Chicken pox	vomiting	vomiting	fatigue	high_fever	headache	nausea	abdominal_pain	I
freq	10	46	56	46	30	24	21	22	

```
df.isnull().sum()
```

	0
Disease	0
Symptom_0	0
Symptom_1	0
Symptom_2	0
Symptom_3	32
Symptom_4	79
Symptom_5	127
Symptom_6	155
Symptom_7	173
Symptom_8	193
Symptom_9	203
Symptom_10	245
Symptom_11	266
Symptom_12	283
Symptom_13	294
Symptom_14	295
Symptom_15	303
Symptom_16	312

dtype: int64

```
df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
df.fillna(df.median(numeric_only=True), inplace=True)
```

```
df.dropna(inplace=True)
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.drop_duplicates(inplace=True)
```

```
print("actual columns in df:",df.columns.tolist())
df.columns = df.columns.str.strip()
symptom_columns = [col for col in df.columns if col.startswith('Symptom_')]
for col in symptom_columns:
    df[col] = df[col].str.lower()
```

```
actual columns in df: ['Disease', 'Symptom_0', 'Symptom_1', 'Symptom_2', 'Symptom_3', 'Symptom_4',
```

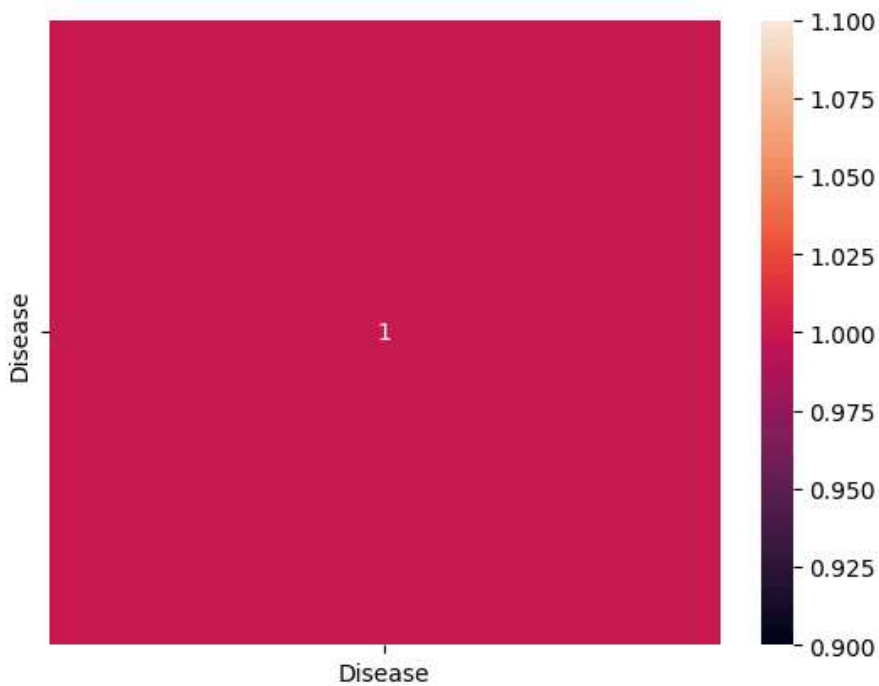
Start coding or generate with AI.

```
import re
symptom_columns = [col for col in df.columns if col.startswith('Symptom_')]
for col in symptom_columns:
    df[col] = df[col].apply(lambda x: re.sub('[^a-zA-Z ]','', str(x)))
```

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Disease'] = le.fit_transform(df['Disease'])
```

```
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

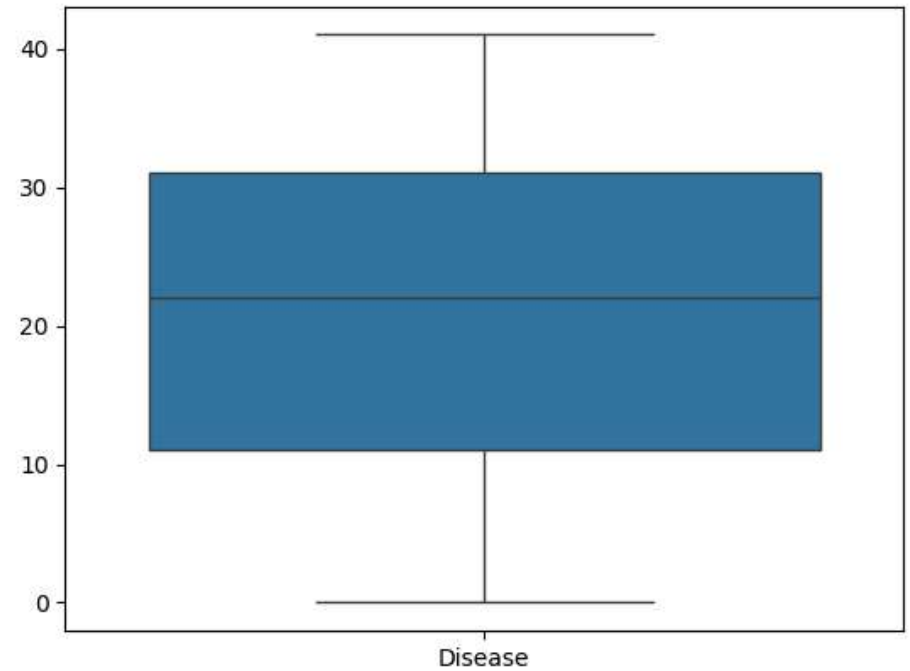
```
<Axes: >
```




```
numerical_df = df.select_dtypes(include=['number'])
Q1 = numerical_df.quantile(0.25)
Q3 = numerical_df.quantile(0.75)
```

```
sns.boxplot(data=df)
```

```
<Axes: >
```




```
X = df.drop('Disease', axis=1)
y = df['Disease']
```

```
from sklearn.model_selection import train_test_split
import pandas as pd

X = df.drop("Disease", axis=1)
y = df["Disease"]

print("Total rows:", len(df))

if len(df) <= 1:
    print("Warning: DataFrame has 1 or fewer rows. Cannot perform a meaningful train-test split.'
    print("Assigning all data to training set and creating empty test sets to avoid error.")
    X_train, y_train = X, y
    X_test = pd.DataFrame(columns=X.columns)
    y_test = pd.Series(dtype=y.dtype)
else:
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.5, random_state=42
    )

print(f"X_train shape: {X_train.shape}")
print(f"X_test shape: {X_test.shape}")
```

```
Total rows: 313
X_train shape: (156, 17)
X_test shape: (157, 17)
```

```
from sklearn.ensemble import RandomForestClassifier
import pandas as pd

# Identify categorical columns for one-hot encoding (all Symptom_ columns)
symptom_columns = [col for col in X_train.columns if col.startswith('Symptom_')]
```

```
# Apply one-hot encoding to X_train and X_test
X_train_encoded = pd.get_dummies(X_train, columns=symptom_columns, prefix=symptom_columns)
X_test_encoded = pd.get_dummies(X_test, columns=symptom_columns, prefix=symptom_columns)

# Align columns - this is crucial to ensure both datasets have the same columns
# If a symptom appears in train but not test, or vice-versa, get_dummies will produce different c
# Reindex ensures both dataframes have the union of all columns, filling with 0 if a column is mi
all_columns = list(set(X_train_encoded.columns) | set(X_test_encoded.columns))
X_train_encoded = X_train_encoded.reindex(columns=all_columns, fill_value=0)
X_test_encoded = X_test_encoded.reindex(columns=all_columns, fill_value=0)

model = RandomForestClassifier(random_state=42)

model.fit(X_train_encoded, y_train)
```

```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

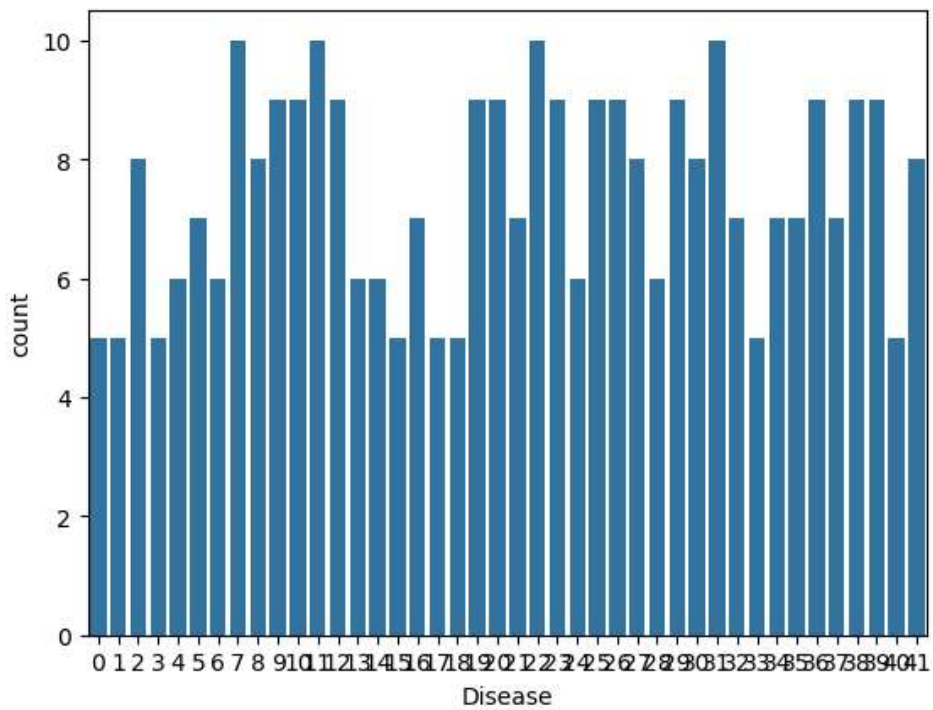
```
y_pred = model.predict(X_test_encoded)
```

```
from sklearn.metrics import accuracy_score, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

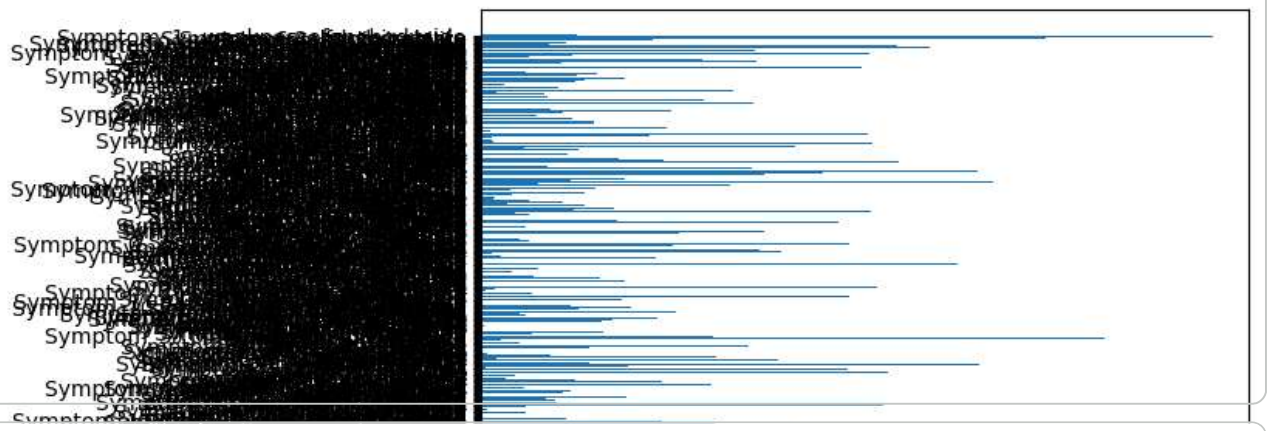
```
Accuracy: 0.8343949044585988
              precision    recall  f1-score   support

           0       0.40      1.00      0.57         2
           1       0.80      1.00      0.89         4
           2       0.80      1.00      0.89         4
           3       1.00      0.67      0.80         3
           4       1.00      1.00      1.00         2
           5       1.00      1.00      1.00         3
           6       1.00      1.00      1.00         2
           7       1.00      1.00      1.00         3
           8       1.00      1.00      1.00         4
           9       0.83      1.00      0.91         5
          10       0.75      0.43      0.55         7
          11       0.83      1.00      0.91         5
          12       1.00      1.00      1.00         4
          13       1.00      1.00      1.00         2
          14       0.67      1.00      0.80         2
          15       1.00      0.50      0.67         4
          16       0.00      0.00      0.00         7
          17       1.00      1.00      1.00         2
          18       0.43      1.00      0.60         3
          19       1.00      0.67      0.80         3
          20       1.00      0.86      0.92         7
          21       1.00      1.00      1.00         3
          22       0.57      0.80      0.67         5
          23       0.75      1.00      0.86         3
          24       1.00      0.75      0.86         4
          25       1.00      1.00      1.00         6
          26       1.00      0.83      0.91         6
          27       0.50      0.50      0.50         2
          28       0.67      0.67      0.67         3
          29       1.00      1.00      1.00         3
          30       1.00      1.00      1.00         5
          31       0.80      1.00      0.89         4
          32       1.00      1.00      1.00         4
```

```
           33       1.00      0.33      0.50         3
           34       0.67      1.00      0.80         2
           35       0.38      1.00      0.55         3
           36       1.00      1.00      1.00         2
           37       1.00      1.00      1.00         3
           38       1.00      0.83      0.91         6
           39       1.00      0.86      0.92         7
           40       1.00      0.50      0.67         2
           41       1.00      1.00      1.00         3

    accuracy                           0.83       157
   macro avg       0.85      0.86      0.83       157
weighted avg       0.85      0.83      0.82       157

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWa
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWa
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWa
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

```
sns.countplot(x=y)
plt.show()
```




```
importance = model.feature_importances_

plt.barh(X_train_encoded.columns, importance)
plt.show()
```



```
import pandas as pd
import re

# Define the symptom columns, which are available from the notebook state
symptom_columns = ['Symptom_0', 'Symptom_1', 'Symptom_2', 'Symptom_3', 'Symptom_4', 'Symptom_5',

# Create a sample DataFrame with symptoms for prediction.
# Replace the symptom strings below with the actual symptoms you want to predict.
# Use 'nan' for symptoms that are not present in the sample.
# Example uses symptoms from the first row of the original dataset:
new_sample_data = pd.DataFrame([[
    'musclewasting', 'patchesinthroat', 'highfever', 'extramaritalcontacts',
    'nan', 'nan', 'nan', 'nan', 'nan', 'nan', 'nan', 'nan', 'nan', 'nan', 'nan', 'nan', 'nan'
]], columns=symptom_columns)

# Apply cleaning steps similar to how the training data was processed
for col in symptom_columns:
    # Ensure value is string before calling .lower()
    new_sample_data[col] = new_sample_data[col].astype(str).str.lower()
    new_sample_data[col] = new_sample_data[col].apply(lambda x: re.sub('[^a-zA-Z ]','', str(x)))

# One-hot encode the new sample data
sample_encoded = pd.get_dummies(new_sample_data, columns=symptom_columns, prefix=symptom_columns)

# Align the columns of the encoded sample with the columns the model was trained on.
# `all_columns` variable is available from the kernel state after model training.
sample_encoded = sample_encoded.reindex(columns=all_columns, fill_value=0)

# Now make the prediction
prediction = model.predict(sample_encoded)
```